

120 Days of Quantitative Finance

Month 3: Machine Learning for Finance

Week 10: Advanced ML Models

Day 69: Feature Importance in Trading

1 Introduction

Machine learning models in finance often use a large number of input features derived from market data. These features may include returns, volatility indicators, momentum signals, volume measures, and macroeconomic variables.

However, not all features contribute equally to predictions. Some features contain strong predictive information, while others may introduce noise or lead to overfitting.

Feature importance refers to techniques used to quantify how much each input variable contributes to the predictive performance of a model.

Understanding feature importance helps traders:

- Identify meaningful market signals
- Reduce model complexity
- Improve model interpretability
- Avoid overfitting

2 Feature Representation

Suppose we have a dataset with n observations and p features:

$$X = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1p} \\ x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{np} \end{bmatrix}$$

and a target variable representing returns:

$$y = (y_1, y_2, \dots, y_n)$$

A machine learning model learns a function:

$$\hat{y} = f(X)$$

Feature importance attempts to measure how strongly each feature x_j influences the prediction.

3 Feature Importance in Linear Models

For linear regression models:

$$y = \beta_0 + \sum_{j=1}^p \beta_j x_j + \epsilon$$

The coefficients β_j measure the contribution of each feature.

A larger magnitude coefficient indicates a stronger relationship between the feature and the target variable.

However, coefficients must be interpreted carefully if features are on different scales or highly correlated.

4 Feature Importance in Tree-Based Models

Tree-based models such as Random Forests and Gradient Boosting estimate feature importance using **impurity reduction**.

During training, decision trees split data based on features that reduce prediction error. The importance of feature j is measured as the total reduction in impurity contributed by that feature.

Formally:

$$FI_j = \sum_{t \in T_j} \Delta I_t$$

where

- FI_j is the feature importance of feature j
- T_j represents all tree splits using feature j
- ΔI_t is the reduction in impurity at node t

Features that frequently reduce prediction error will receive higher importance scores.

5 Permutation Feature Importance

Another popular approach is **permutation importance**.

The idea is simple:

1. Measure the model's baseline performance.
2. Randomly shuffle one feature.
3. Measure the new model performance.
4. The drop in performance represents the importance of that feature.

If shuffling a feature significantly reduces prediction accuracy, the model depends strongly on that feature.

Mathematically:

$$FI_j = Performance_{baseline} - Performance_{permuted}$$

6 Feature Importance in Trading Strategies

In financial markets, common predictive features include:

- Momentum indicators

- Volatility measures
- Moving averages
- Volume signals
- Macro variables

Feature importance analysis can reveal which signals actually drive predictive models. For example, a model predicting returns may discover:

Feature	Importance
Momentum (20-day)	High
Volatility	Medium
Volume change	Low

This information helps researchers focus on the most informative signals.

7 Challenges in Financial Feature Importance

Feature importance analysis in finance has several challenges:

- Financial features are often highly correlated
- Market regimes can change over time
- Importance may vary across different models
- Noise in financial data can distort interpretations

Therefore, feature importance should always be validated with out-of-sample testing.

8 Conclusion

Feature importance is a powerful tool for interpreting machine learning models in trading.

By identifying the most predictive signals, traders can build simpler, more robust models and gain deeper insight into the drivers of market behavior.

Understanding which features matter most is a crucial step toward building reliable quantitative trading strategies.